Oracle Utilities Customer Care and Billing v25.10HOT

Product Fix Design

Bug 38410113

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Oracle Utilities Customer Care and Billing Product Fix Design

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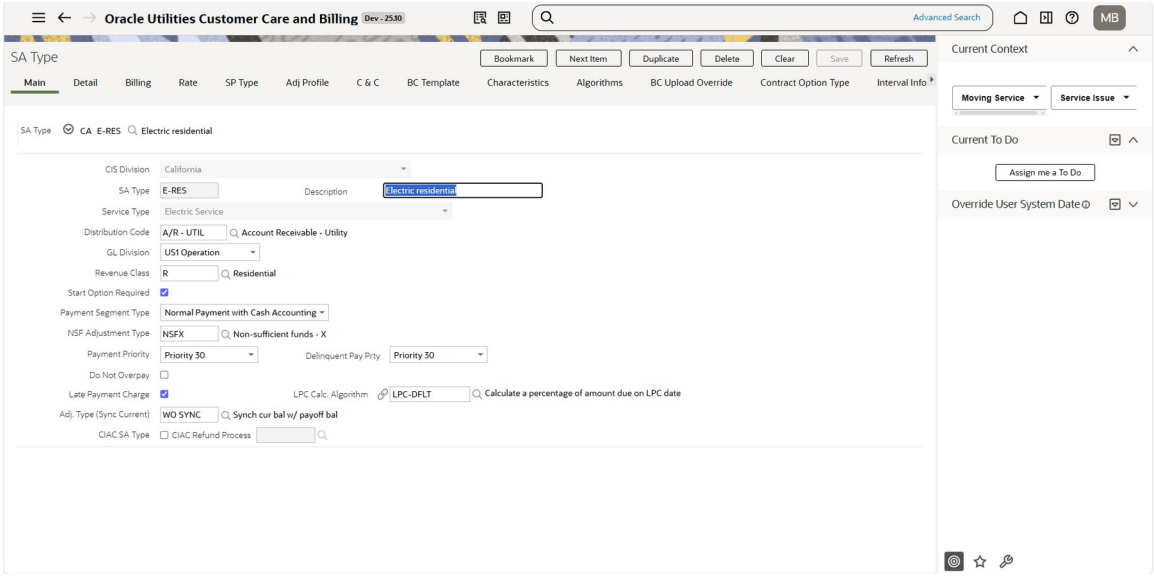
Reported Issue

Description

An SQL error code ***"999999994 occurred in module CIPBDCPN:XQ030 SELECT"*** is encountered when generating a bill for an account whose Service Agreement Type exists in multiple CIS Divisions and the "Override Consumption Period" feature configuration option is enabled. The failure occurs during execution of **CIPBDCPN**, which invokes **CIPBUSPX** to validate Override Consumption Period eligibility.

Steps To Recreate

Data Setup



SA Type E-RES in CIS Division CA

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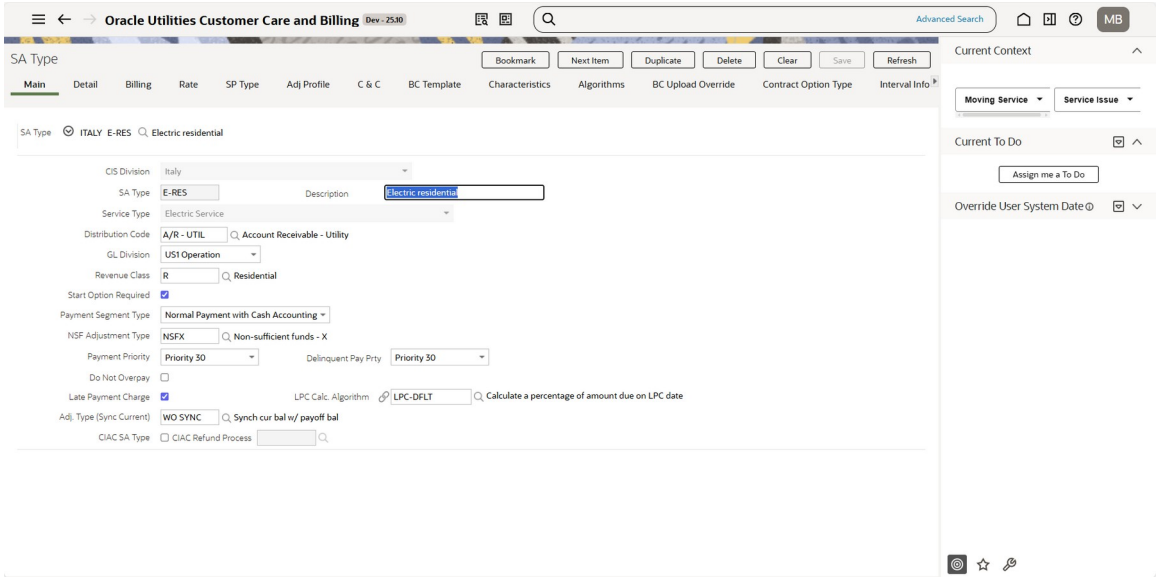
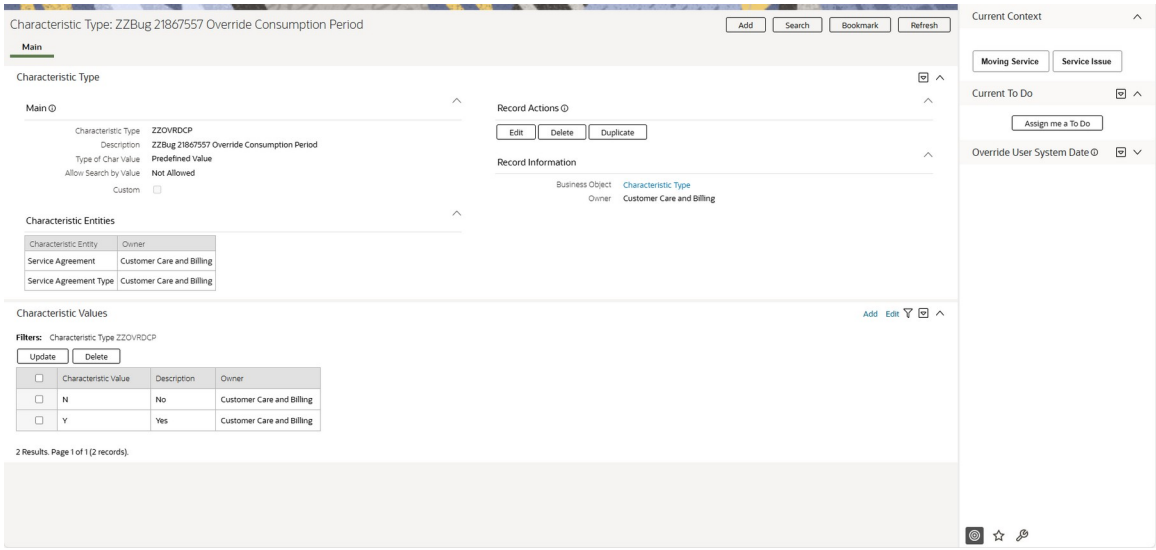


SA Type E-RES in CIS Division ITALY

Create two SA Types with the same SA Type code across different CIS Divisions



Characteristic Type setup for Override Consumption Period

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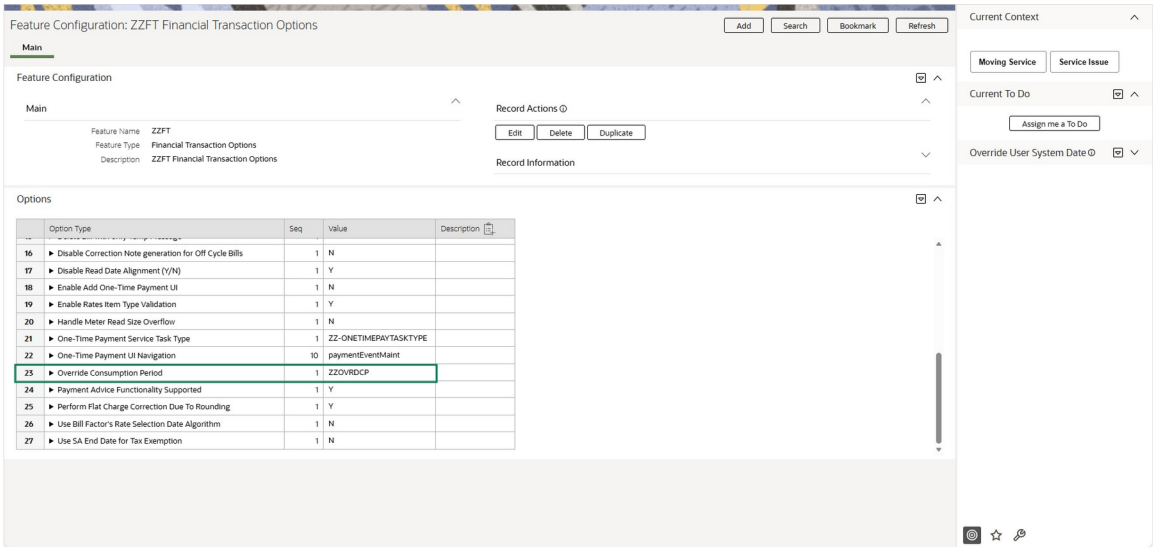
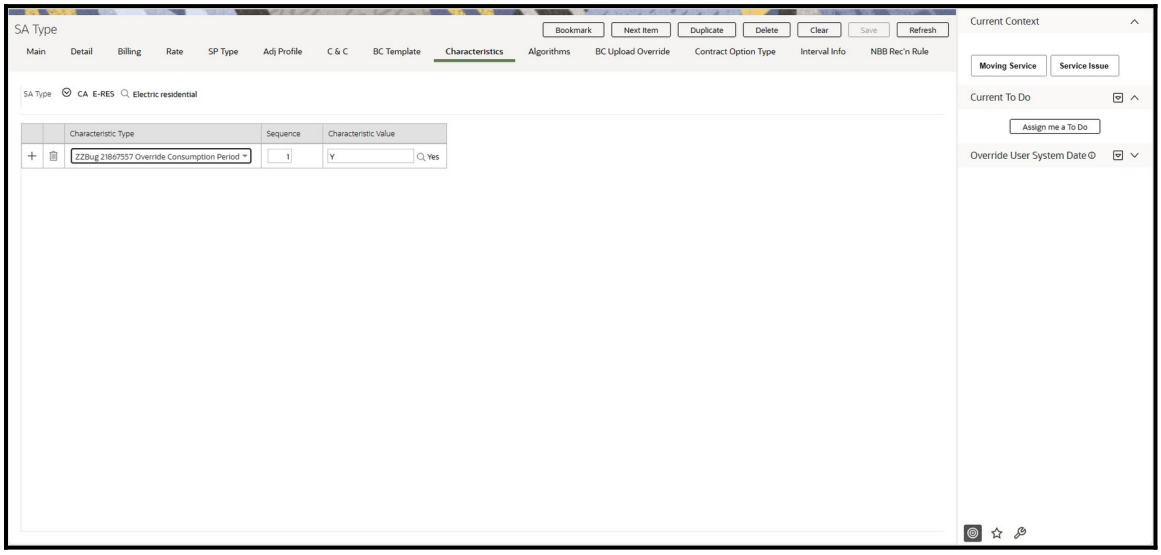


Override Consumption Period feature configuration using Characteristic Type ZZOVRDCP



Add Override Consumption Period characteristic to SA Type (CIS Division = CA)

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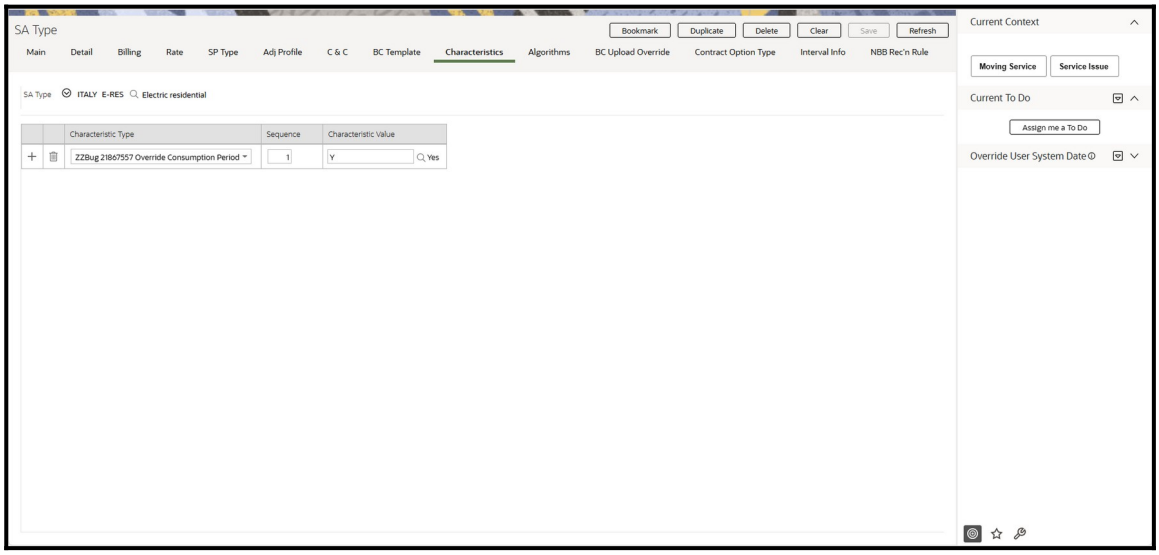
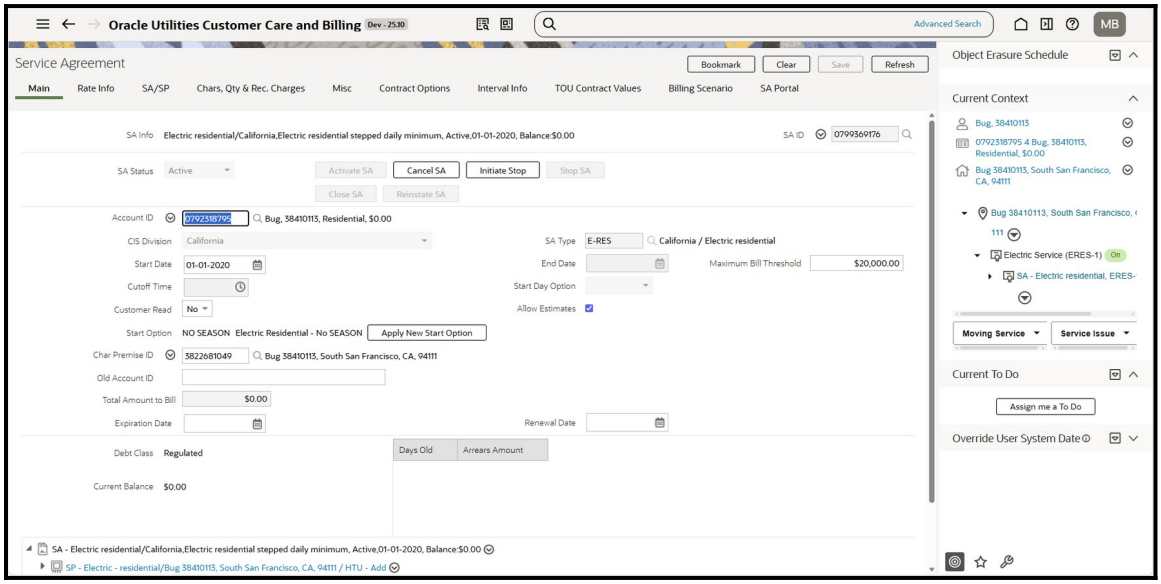


Add Override Consumption Period characteristic to SA Type (CIS Division = ITALY)



Create an account using the configured SA Type

Replication Steps

- Generate a bill for the account that uses an SA Type configured in multiple CIS Divisions with the Override Consumption Period feature enabled. Notice that the billing process fails with ***SQL Error 999999994*** raised from ***CIPBDCPN:XQ030 SELECT***.

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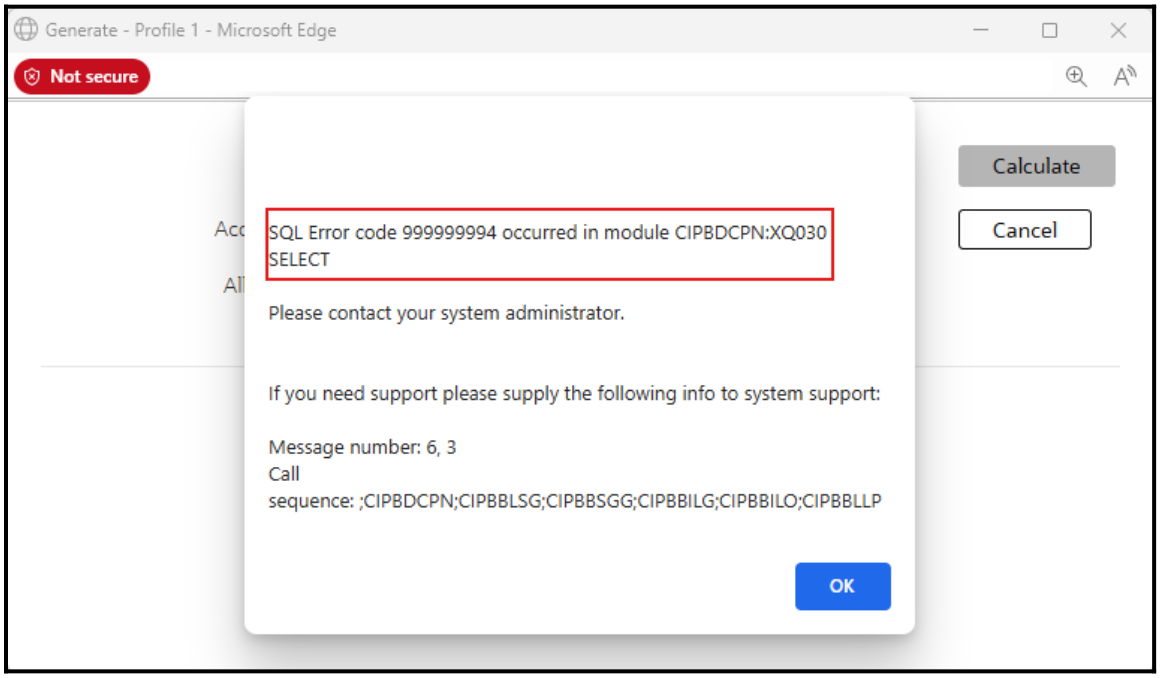

Error dialog showing failure in CIPBDCPN (XQ030 SELECT) when generating bill

Solution Design

- Update the subsequent program logic to reference the returned count result instead of directly relying on the existence of a single SQL result row. When the count is greater than zero, proceed with Override Consumption Period logic; otherwise bypass it and use default billing consumption logic.
- Modify the translated COBOL programs ***CIPBUSPX*** and ***CIPBDCPN*** to replace the direct single-row lookup against ***CI_SA_TYPE_CHAR*** with a ***COUNT(*)***-based check. This ensures that scenarios with multiple SA Types across CIS Divisions do not trigger a SQL error due to multiple row retrievals.

Summary

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N

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Does the fix involve DB Changes?	N
Is there an impact on any user interface or any other on-line behavior?	**Y**
Is there an impact on any batch process?	N
Is there an impact on CM – Custom Modification processing?	N
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

Translated Java Routines

ProgramCIPBDCPN – Determine Consumption Period for non-billable charge Bill Segments (Existing)

Regenerate the Gen file to get the new SQL changes.

Business Rules

The SQL used to validate whether a Service Agreement Type supports Override Consumption Period was updated to return a **row count** instead of a single value. This prevents SQL errors caused by multiple matching rows on ***CI_SA_TYPE_CHAR*** when the same SA Type exists across multiple CIS Divisions. The logic now treats ***COUNT(*) > 0*** as a successful match, allowing the program to continue without failure while still enabling Override Consumption.

SQL Changes (Oracle)

S-CHKOCCT – Check Override Consumption Characteristic Type

SQL Before Change

```
SELECT
   'X'
FROM
   CI_SA_TYPE_CHAR
WHERE SA_TYPE_CD = &IN.SA-TYPE-CD
AND CHAR_TYPE_CD = &IN.CHAR_TYPE_CD
```

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Execution Plan Before Change

OPERATION	OBJECT_NAME	OPTIONS	CARDINALITY	COST
SELECT STATEMENT			1	4
INDEX	CISADM.XC323P0	FAST FULL SCAN	1	4

SQL After Change

```
SELECT
    COUNT(*) AS CNT
FROM
    CI_SA_TYPE_CHAR STC
WHERE STC.SA_TYPE_CD   = &IN.SA-TYPE-CD
AND   STC.CHAR_TYPE_CD = &IN.CHAR-TYPE-CD
```

Execution Plan After Change

Explain Plan

SQL | 0.563 seconds

OPERATION	OBJECT_NAME	OPTIONS	CARDINALITY
SELECT STATEMENT			1
SORT		AGGREGATE	1
INDEX	CISADM.XC323P0	FAST FULL SCAN	1
Filter Predicates			
Other XML			

ProgramCIPBUSPX – Get Consumption Algorithm. (EXISTING)

Updated to Non-Regeneratable, will manually add the SQL changes to feed the CIS Division filter.

Business Rules

The SQL used to validate whether a Service Agreement Type supports Override Consumption Period was updated to return a **row count** instead of a single value. This prevents SQL errors caused by multiple matching rows on ***CI_SA_TYPE_CHAR*** when the same SA Type exists across multiple CIS Divisions. The logic now treats ***COUNT(*) > 0*** as a successful match, allowing the program to continue without failure while still enabling Override Consumption.

SQL Changes (Oracle)

S-CHKOCCT – Check Override Consumption Characteristic Type

SQL Before Change

```
SELECT
    'X'
FROM
    CI_SA_TYPE_CHAR
WHERE SA_TYPE_CD = &IN.SA-TYPE-CD
AND CHAR_TYPE_CD = &IN.CHAR_TYPE_CD
```


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Execution Plan Before Change

OPERATION	OBJECT_NAME	OPTIONS	CARDINALITY	COST
SELECT STATEMENT			1	4
INDEX	CISADM.XC323P0	FAST FULL SCAN	1	4

SQL After Change

```
SELECT
    COUNT(*) AS CNT
FROM
    CI_SA_TYPE_CHAR STC
WHERE STC.SA_TYPE_CD   = &IN.SA-TYPE-CD
AND   STC.CHAR_TYPE_CD = &IN.CHAR-TYPE-CD
```

Execution Plan After Change

Explain Plan

SQL | 0.563 seconds

OPERATION	OBJECT_NAME	OPTIONS	CARDINALITY
SELECT STATEMENT			1
SORT		AGGREGATE	1
INDEX	CISADM.XC323P0	FAST FULL SCAN	1
Filter Predicates			
Other XML			

ProgramCIPBDCPN_Impl – Determine Consumption Period for non-billable charge Bill Segments (Existing)

Determines the Bill Segment Consumption Period and applies Override Consumption Period when enabled in Feature Configuration and the Service Agreement's SA Type contains the configured characteristic. The change ensures that instead of selecting a single characteristic row, the program now counts matching rows and treats ***count > 0*** as a positive match, preventing multiple-row SQL errors and allowing Override Consumption Period to work when multiple SA Types exist across CIS Divisions.

Business Rules

perform_GC030_CHECK_OVRD_CONSUMP_PRD_FC()

- Check if Override Consumption Period is configured in Feature Configuration
- Set External System Type = ***C1FT***
  - Set Option Type = Override Consumption Period
  - Retrieve Feature Configuration option
- If Feature Configuration option exists
  - Initialize lookup input values

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  - Set SA Type Code from the Service Agreement
  - Set Characteristic Type Code from the Feature Configuration
  - Execute ***CHKOCCT*** SQL to verify matching SA Type characteristic
  - SQL returns the **count of matching rows**
- **If count > 0**
  - Set Override Consumption Period Switch = **true**
- Else
  - Override Consumption Period Switch remains **false**

ProgramCIPBUSPX_Impl – Get Consumption Algorithm (Change)

Calculates the Consumption Period for Service Cycle-based billing and applies Override Consumption Period when enabled in Feature Configuration and the Service Agreement's SA Type contains the configured characteristic. The change introduces the same count-based check ***(count > 0)*** for characteristic lookup to avoid multiple-row SQL errors and correctly enable Override Consumption Period in cases where multiple SA Types exist across CIS Divisions.

Business Rules

perform_MA090_CHECK_OVRD_CONSUMP_PRD_FC()

- Check if Override Consumption Period is configured in Feature Configuration
- Set External System Type = ***C1FT***
  - Set Option Type = Override Consumption Period
  - Retrieve Feature Configuration option
- If Feature Configuration option exists
  - Initialize lookup input values
  - Set SA Type Code from the Service Agreement
  - Set Characteristic Type Code from the Feature Configuration
  - Execute ***CHKOCCT*** SQL to verify matching SA Type characteristic
  - SQL returns the **count of matching rows**
- **If count > 0**
  - Set Override Consumption Period Switch = **true**
- Else
  - Override Consumption Period Switch remains **false**

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Required Data Changes

Data Clean-Up / Fixes

- Not Applicable

Recommended Test Cases

Test Case 1: Reported Issue - Generate Bill with Override Consumption

- Refer to Replication Steps. Generate a Bill and verify that the Bill is generated successfully and there's no SQL error encountered. Also, verify that the Bill segments are successfully generated without any error.
- **Expected Result:** The Generated Bill segments are successfully generated without any error.

Test Case 2: Generate Bill with Override Consumption and Override Consumption Service Agreement

- Refer to Replication Steps. Create a Person and Account with an activated Service Agreement for Override Consumption, Generate a Bill and berify that the Bill is generated successfully and there's no SQL error encountered. Also, verify if the Consumption period was changed, "-1 day" should have been applied.
- **Expected Result:** The Generated Bill segments are successfully generated without any error. Consumption Period changes applied "-1" day.

Test Case 3: Generate Bill — Single Row Returned

- Refer to Replication Steps. Validate that when only **one** SA Type has the Override Consumption Period characteristic configured, the SQL returns a single matching row and the bill processing works without any SQL error.
- **Expected Result:** The Generated Bill segments are successfully generated without any error.

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Test Case 4: Generate Bill — No Rows Returned

- Refer to Replication Steps. Validate that when **no** SA Type has the Override Consumption Period characteristic configured, the SQL returns zero rows and billing still operates without error.
- **Expected Result:** The Generated Bill segments are successfully generated without any error.

Test Case 5: Regression Testing

- Perform regression testing on old functionality. Rate Version 1 & 2 and no Override Consumption Period Feature Configuration defined.
- **Expected Result:** The bill should generate successfully without errors and without invoking the override consumption logic since no Feature Configuration is defined. The system should use the original consumption period calculation rules for Rate Version 1 & 2, and all bill segments, dates, and values should remain consistent with baseline behavior, confirming that no unintended changes were introduced by the fix.

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